

$$\begin{bmatrix} \textcircled{1} & 2 & -1 & 4 \\ 0 & \textcircled{2} & 0 & 3 \\ 0 & 0 & \textcircled{1} & -2 \end{bmatrix}$$

ref ✓
rref X

$$\begin{bmatrix} \textcircled{1} & 2 & -1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & \textcircled{1} & 2 & -4 \end{bmatrix}$$

ref X
rref X

لازم يكون تحت

$$\begin{bmatrix} \textcircled{1} & -5 & 2 & -1 & 3 \\ 0 & 0 & \textcircled{1} & 3 & -2 \\ 0 & 0 & 0 & \textcircled{1} & 4 \\ 0 & 0 & 0 & 0 & \textcircled{1} \end{bmatrix}$$

ref ✓
rref X

$$\begin{bmatrix} \textcircled{1} & 0 & 0 & -1 \\ 0 & \textcircled{1} & 0 & 2 \\ 0 & 0 & \textcircled{1} & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

ref ✓
rref ✓

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & \textcircled{2} & 1 & -1 \\ 0 & 0 & 1 & -3 \end{bmatrix}$$

ref X
rref X

لنفس اول يقابل في الصف واحد

$$-12 + 8 = -4$$

Notes

$$A + B = B + A$$

$$A + (B + C) = (A + B) + C$$

$$A + O = O + A = A$$

$$A - A = -A + A = O$$

$$\begin{array}{ccc} A & B & AB \\ m \times n & n \times p & = m \times p \end{array}$$

Part 9
matrix multiplication

$$A = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix}$$

$$3 \times 2$$

$$B = \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix}$$

$$2 \times 2$$

$$\begin{bmatrix} -9 & 1 \\ -15 & 10 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 4 & 6 \\ 2 & -1 \end{bmatrix}$$

$$AB = \begin{bmatrix} (2 \times 4 + 1 \times 2) & (2 \times 6 + 1 \times -1) \\ (3 \times 4 + 0 \times 2) & (3 \times 6 + 0 \times -1) \end{bmatrix} = \begin{bmatrix} 10 & 11 \\ 12 & 18 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -1 \\ 2 & -3 \end{bmatrix}$$

$$A^2 = A \cdot A = \begin{bmatrix} 1 & -1 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & -3 \end{bmatrix} \\ = \begin{bmatrix} -1 & 2 \\ -4 & 7 \end{bmatrix}$$

Note:

$$A(BC) = (AB)C$$

$$A(B+C) = AB+AC$$

$$(A+B)C = AC+BC$$

$$AB \neq BA$$

[true or false] —

if

$$AC = BC, \text{ then } A = B$$

$$AC = BC$$

$$AC - BC = 0$$

$$(A - B)C = 0$$

$$AC = 0$$

then A may be not equal to

False

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$$(A^T)$$

Part 10

The Transpose
of a
Matrix

$$A = \begin{bmatrix} 3 & 4 \\ -2 & 5 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 3 & -2 \\ 4 & 5 \end{bmatrix}$$

$$B = [3 \quad -2 \quad 5]$$

$$B^T = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 3 & 4 & 5 & 6 \\ 2 & -2 & 2 & -2 \\ 7 & 8 & 9 & 0 \end{bmatrix}$$

$$C^T = \begin{bmatrix} 1 & 3 & 2 & 7 \\ -1 & 4 & -2 & 8 \\ 1 & 5 & 2 & 9 \\ -1 & 6 & -2 & 0 \end{bmatrix}$$

Determinant
a Transpose

$$\det(A) = \det(A^T)$$

$$A = \begin{bmatrix} 3 & 1 & -2 \\ 2 & 0 & 0 \\ -4 & -1 & 5 \end{bmatrix}$$

$$|A| = 2(-1)^3 \begin{vmatrix} 1 & -2 \\ -1 & 5 \end{vmatrix}$$

$$1 \times 5 - 2 = 3$$

$$2 \times -1 \times 3 = -6$$

$$A^T = \begin{bmatrix} 3 & 2 & -4 \\ 2 & 0 & -1 \\ -2 & 0 & 5 \end{bmatrix}$$

$$|A^T| = 2(-1)^3 \begin{vmatrix} 1 & -1 \\ -2 & 5 \end{vmatrix}$$

$$(2) \times -1 \times (3)$$

$$= -6$$

$$|A| = |A^T|$$

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Does the set of vectors

$$V_1 = (1, 1, 2), V_2 = (1, 0, 1)$$

$$V_3 = (2, 1, 3)$$

span to $\mathbb{R}^3$?

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$c_1 + c_2 + 2c_3 = w_1$$

$$c_1 + c_3 = w_2$$

$$2c_1 + c_2 + 3c_3 = w_3$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix} \neq 0 \rightarrow \text{span}$$

Determine whether each vector can be written as a linear combination of the vectors in

$$S = \{(2, -1, 3), (5, 0, 4)\}$$

(a) $w = (1, 1, -1)$

(b) $v = (8, \frac{1}{4}, \frac{27}{4})$

(c) $(1, -8, 12)$

(d) $\vec{z} = (-1, -2, 2)$

$$x(1, 1, -1) = c_1(2, -1, 3) + c_2(5, 0, 4)$$

$$\left(8, -\frac{1}{4}, \frac{27}{4}\right)$$

$$2c_1 + 5c_2 = 1 \quad (1)$$

$$c_1 = -1 \quad (2)$$

$$3c_1 + 4c_2 = -1 \quad (3)$$

يُبقَى مِنْ فَتَوَى طَبَّاكٍ مَشْهُورٍ

فَلْيُحَقِّقُوا الْمَادَّةَ الْكَامِلَةَ

$$(1, -8, 12) = c_1(2, -1, 3) + c_2(5, 0, 4)$$

$$-16 +$$

$$2c_1 + 5c_2 = 1$$

$$-c_1 = -8$$

$$3c_1 + 4c_2 = 12$$

$$3 \times -8 + 4c_2 = 12$$

$$-24 + 4c_2 = 12$$

$$4c_2 = 36$$

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Linear combination

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$$X = C_1 V_1 + C_2 V_2 + \dots + C_K V_K$$

Exp

Vector $x = (-1, -2, -2)$ is a linear combination of $u = (0, 1, 4)$, $v = (-1, 1, 2)$, and $w = (3, 1, 2)$

$$\text{Since } x = u - 2v - w$$

$$\begin{matrix} u & & -2v & & -w \\ \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix} & + & \begin{bmatrix} 2 \\ -2 \\ -4 \end{bmatrix} & + & \begin{bmatrix} -3 \\ -1 \\ -2 \end{bmatrix} \end{matrix}$$

$$= \begin{bmatrix} -1 \\ -2 \\ -2 \end{bmatrix} X$$

Example 1

show that the set $V = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \mid x > y \right\}$, together with the standard operations of vector addition and scalar multiplication is not a vector space.

$$-\begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ -1 \end{bmatrix} \notin V$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \notin V$$

1) $\vec{0} \notin V$ $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \not> 0$ not a vector space

$$2) u = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \in V \leadsto -1(u) = \begin{pmatrix} -2 \\ -1 \end{pmatrix} \in V$$

$-2 < -1$

$V = \mathbb{R}^3$ together with the two operations defined as:

$$\forall x, y, z \in V, \text{ and } \forall r \in \mathbb{R}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \oplus \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} x + y' \\ y + z' \\ z + x' \end{bmatrix} \text{ and } r \otimes \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} rx \\ ry \\ rz \end{bmatrix}$$

$\oplus$
vector addition

$\otimes$ or $\odot$
scalar multiplication

is a vector space

is $(V, \oplus, \otimes)$ a vector space?

$$\vec{0} \in V \quad \vec{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\vec{u} + \vec{0} = \vec{u} \quad ??$$

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} \oplus \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \vec{u}$$

$$\vec{u} + \vec{v}$$

